



NOTRE DAME UNIVERSITY

BANGLADESH

Computer Graphics Assignment-04

Course Code: CSE-4204

Course Title: Computer Graphics

Lab Task Topic: 2D Transformation

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Question: Suppose, there are 3 points $A(3,2)$, $B(6,2)$ & $C(6,5)$ on a 2D plane. Apply composite transformation if it is possible in the given scenario. Show that X direction shear where $Sh_x = 4$. Then scale the object uniformly where $S_y = 0.5$ after reflecting along y axis.

Solution: Given, $A(3,2)$, $B(6,2)$, $C(6,5)$, $Sh_x = 4$

$$\begin{aligned} A'_x &= A_x + Sh_x * A_y \\ &= 3 + 4 \cdot 2 = 11 \end{aligned}$$

$$A'_y = A_{y_{old}} = 2$$

$$\begin{aligned} B'_x &= B_x + Sh_x * B_y \\ &= 6 + 4 \cdot 2 = 14 \end{aligned}$$

$$B'_y = B_{y_{old}} = 2$$

$$\begin{aligned} C'_x &= C_x + Sh_x * C_y \\ &= 6 + 4 \cdot 5 = 26 \end{aligned}$$

$$C'_y = C_{y_{old}} = 5$$

Reflection on Y-axis, $x = -x$ & $y = y$

$$A'(11,2) \longrightarrow A''(-11,2)$$

$$B'(14,2) \longrightarrow B''(-14,2)$$

$$C'(26,5) \longrightarrow C''(-26,5)$$

Now scaling the object $S_y = 0.5$ as $(x, 0.5y)$

$$\therefore A'(11,2) \longrightarrow A''(-11, 0.5 \times 2) = A'''(-11, 1)$$

$$\therefore B'(14,2) \longrightarrow B''(-14, 0.5 \times 2) = B'''(-14, 1)$$

$$\therefore C'(26,5) \longrightarrow C''(-26, 0.5 \times 5) = C'''(-26, 2.5)$$

(Ans)